PHYS 143 – Mechanics and Thermal Physics

Newton's Laws and Forces

How do forces determine how an object moves?

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By the end of this week, you should be able to:

1. Identify forces and represent them using free-body diagrams.
2. Apply Newton's laws to equilibrium and accelerating systems.
3. Model normal force, tension and friction in physical systems.
4. Solve connected-object, inclined-plane and circular-dynamics problems.

Central question

How do forces determine how an object moves?

The thread of the week

interaction → **force** → **net force** → $\sum \vec{F} = m\vec{a}$ → **motion.** Draw the free-body diagram first; then write $\sum \vec{F} = m\vec{a}$ by components.

Two contacts: A (Tuesday) **REPRESENT** forces and the first and third laws, then begin to **ANALYSE** with the second law for one body; **B** (Wednesday) **ANALYSE** connected objects and friction, then **APPLY** the laws to slopes, circular motion and an integrated problem.

Reading: Young and Freedman, Chapter 4, §4.1–4.6 and Chapter 5, §5.1–5.4. **Labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Not marked. Answer individually, then check the answers.

1. A $50\,\mathrm{N}$ force acts 30° above the $+x$ axis. Find its components.

   $F_x = 50\cos 30^\circ = 43.3\,\mathrm{N}$, $F_y = 50\sin 30^\circ = 25\,\mathrm{N}$.

2. A car goes from rest to $20\,\mathrm{m/s}$ in $8.0\,\mathrm{s}$. Find its acceleration.

   $a = 20/8.0 = 2.5\,\mathrm{m/s^2}$.

3. Forces of $3.0\,\mathrm{N}$ east and $4.0\,\mathrm{N}$ north act on a body. Find the magnitude and direction of the resultant.

   $R = \sqrt{3^2+4^2} = 5.0\,\mathrm{N}$ at $\tan^{-1}(4/3) = 53^\circ$ north of east.

4. A football is kicked upward. What are its velocity and acceleration at the top?

   $v = 0$ (momentarily), $a = 9.8\,\mathrm{m/s^2}$ downward ($a_y = -g$).

5. Solve $T - 20 = 2a$ and $30 - T = 3a$ for T and a.

   Add: $10 = 5a$, so $a = 2.0\,\mathrm{m/s^2}$ and $T = 24\,\mathrm{N}$.

6. An object moves in a circle of radius $2.0\,\mathrm{m}$ at $4.0\,\mathrm{m/s}$. Find its acceleration and its direction.

   $a = v^2/R = 16/2.0 = 8.0\,\mathrm{m/s^2}$, toward the centre.

Calculus is not needed this week. The Week 1 and Week 2 toolkits (components, signs, circular acceleration) are reused throughout.

Contact A: Represent, then Analyse

What forces act, how do we represent them, and how do they fix the acceleration of one body?

Weeks 1 and 2: vectors $\rightarrow \vec{r} \rightarrow \vec{v} \rightarrow \vec{a}$.
How does an object move?

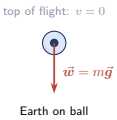
Week 3: interaction $\rightarrow$ force $\rightarrow$ net force $\rightarrow \vec{a}$.
Why does its motion change?

Opening question

A football is kicked straight up. At its highest point, **which forces act on it, and what is the net force?**

Predict first

Poll: (a) none, (b) weight only, (c) weight and an upward "force of the kick", (d) not enough information. Decide before the next click.



(b) Weight only. Only one object interacts with the ball: the Earth, so the force is the **Earth on the ball**, its weight. There is no "force of the kick": the foot stopped acting when the ball left it. The net force is mg downward, so $a = g$ downward (Week 2), even though $v = 0$ for an instant.

Defining the vocabulary of dynamics

CORE

Quantity	What it is	What it measures	SI unit	Representation
Force	an interaction between two objects	the strength and direction of one object's push or pull on another	newton, $1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$	an arrow on the free-body diagram, named by its agent
Net force	the vector sum of all forces on one object	the overall push that determines the acceleration	N	one arrow equal to the vector sum
Mass	a quantitative measure of inertia	how strongly it resists a change of velocity	kg	a number; never drawn as an arrow
Weight	the total gravitational force on an object from all other bodies	how strongly gravity pulls it	N	a downward arrow, $w = mg$

Interpret: the net force fixes the acceleration, and the mass sets how much acceleration a given net force produces.

But what if... a force has no named source? Every force needs an **agent**: we **name** forces first.

Rule: naming a force

Every force is an interaction between **two** objects. Name it by who exerts it on whom:

$$\vec{\boldsymbol{F}}_{\text{A on B}} = \text{the force exerted by } A \text{ on } B$$

- A force is a **vector**: magnitude and direction. Unit: newton, $1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$.
- **Contact** forces: pushes, pulls, surfaces, ropes.
- **Long-range** forces: gravity (and electric and magnetic forces later).

Physical meaning

If you cannot name *the object* that exerts a force, it is not a force. This habit keeps free-body diagrams correct.

Introduce

A footballer kicks a ball lying on the pitch. What can a force do to a body?

Predict first

Name as many effects of a force as you can before the next click.

Define: a force is a push or a pull on a body by another body. It is a vector, with magnitude and direction, measured in newtons (N).

A force can:

(i) set a body at rest in motion: the kick sets the ball moving;
(ii) bring a moving body to rest: the goalkeeper catches the ball;
(iii) change the direction of a moving body: a header redirects the ball;
(iv) speed a moving body up or slow it down: the brakes of a trotro;
(v) change the shape of a body: squeezing a sachet of pure water.

Explain: effects (i) to (iv) are all changes of velocity, that is, *accelerations*. The effect depends on the size *and* the direction of the force. The second law, later in this contact, makes this quantitative: $\sum \vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}}$.

Kinds of force: contact, long-range, and the four fundamental forces

CORE

Introduce. Some forces need touch (a push on a door); others act across empty space (the Earth pulls a ball that has left your hand).

Define:

Contact forces: the bodies touch. Examples: push or pull, normal force, friction, tension.

Long-range (field) forces: act with no contact. Examples: gravity (weight), electric and magnetic forces.

Explain. Every force in nature is one of four fundamental interactions. At the atomic level, normal force, friction and tension are all electromagnetic.

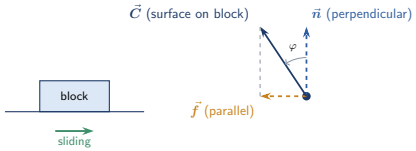
Force	Acts between; does	Range; strength
Gravitational	all masses; attractive; weight, orbits	long; weakest
Electro-magnetic	charged particles; attractive or repulsive; atoms, contact forces	long; $\sim 10^{-2}$
Strong	quarks and nucleons; holds the nucleus together	$10^{-15}\,\mathrm{m}$; strongest, 1
Weak	nuclear decay, such as beta decay	$10^{-18}\,\mathrm{m}$; $\sim 10^{-6}$

Strengths are approximate, relative to the strong force; gravity is about 10^{-38} of it.

Common forces, named by agent

Force	Symbol	Agent on object	Direction
weight	$\vec{w}$	gravity (Earth on object)	toward the Earth, $w = mg$
normal	$\vec{n}$	surface on object	perpendicular to the surface, away from it
tension	$\vec{T}$	rope on object	along the rope, pulling the object
friction	$\vec{f}$	surface on object	parallel to the surface, against the relative sliding
applied	$\vec{F}$	person or machine on object	as applied

All forces are measured in newtons (N). Almost every free-body diagram uses these five. At first write the agent beside each force ("Earth on box"); abbreviate to w, n, T and f once the habit is secure.



Introduce. A rough surface pushes on a sliding block in a tilted direction, not straight up.

Predict: does the surface's push point along the normal?

Contact force

Define: the **contact force** $\vec{C}$ of a surface on a body has two components: the **normal force** $\vec{n}$ (perpendicular to the surface) and the **friction force** $\vec{f}$ (parallel to it).

$C = \sqrt{n^2 + f^2}$, $\tan\varphi = f/n$ (φ from the normal)

Explain. For a sliding block $f = \mu_k n$, so $\tan\varphi = \mu_k$: friction tilts $\vec{C}$ away from the normal by a fixed angle.

Example 7 numbers: $n = 500\,\text{N}$, $f = 200\,\text{N}$

$C = \sqrt{500^2 + 200^2} = 539\,\text{N}$,

$\varphi = \tan^{-1}(0.40) = 21.8^\circ$

Rule: net force

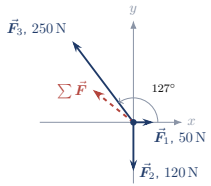
Several forces on one object have the same effect as their **vector sum**:

$$\sum \vec{\boldsymbol{F}} = \vec{\boldsymbol{F}}_1 + \vec{\boldsymbol{F}}_2 + \vec{\boldsymbol{F}}_3 + \cdots \qquad \sum F_x = F_{1x} + F_{2x} + \cdots, \quad \sum F_y = F_{1y} + F_{2y} + \cdots \quad (1)$$

This is Week 1 vector addition by components: **resolve, combine, reconstruct**. Forces at an angle partly cancel, so add by components, not by magnitudes.

Example 1

Three horizontal forces act on a belt, viewed from above: $F_1 = 50\,\mathrm{N}$ along $+x$, $F_2 = 120\,\mathrm{N}$ along $-y$, and $F_3 = 250\,\mathrm{N}$ at 53° to the $-x$ axis (i.e. 127° from $+x$). Find the net force.

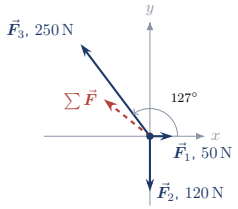


Predict first

From the sketch: which quadrant is the net force in?

Example 1: net force of three forces, solution

CORE



Key idea

Add forces by components: resolve, combine, reconstruct.

IDENTIFY

Given: $F_1 = 50\,\text{N}$ along $+x$, $F_2 = 120\,\text{N}$ along $-y$, $F_3 = 250\,\text{N}$ at $127°$. Find: $\sum \vec{F}$.

SET UP

Vector equation (1): $\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$

Axes: x right, y up, angles from $+x$; $F_x = F\cos\theta$, $F_y = F\sin\theta$.

EXECUTE

Add the x-components:

$\sum F_x = F_1 + 0 + F_3\cos 127° = 50 - 150 = -100\,\text{N}$

Add the y-components:

$\sum F_y = 0 - F_2 + F_3\sin 127° = -120 + 200 = 80\,\text{N}$

Magnitude: $R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = \sqrt{(-100)^2 + (80)^2}\,\text{N}$

Reference angle: $\tan^{-1}(80/100) = 38.7°$ (quadrant II)

$$\boxed{R = 128\,\text{N} \text{ at } \theta = 180° - 38.7° \approx 141° \text{ from } +x}$$

EVALUATE

Quadrant II, as predicted; F_3 dominates; units N. ✓

A handcart thought experiment

A market handcart (a "truck" at Makola) at rest on a level floor is given a small push. On rough concrete it stops quickly; on smooth polished tiles it slides much farther. **What would happen on a perfectly smooth floor?**

Reasoning, line by line:

- After the push, only $\vec{w}$ and $\vec{n}$ act on the cart. They are vertical and cancel; there is no horizontal force.
- So $\sum \vec{F} = 0$, hence $\vec{a} = 0$ and $\vec{v}$ is constant.
- The cart slows on a rough floor only because **friction** acts on it.

Conclusion: no force is needed to *keep* an object moving; a force is needed to *change* its velocity.

Rule: Newton's first law

A body acted on by no net force moves with constant velocity (which may be zero): it stays at rest, or moves in a straight line at constant speed.

$$\sum \vec{F} = 0 \;\Rightarrow\; \vec{a} = 0 \qquad (2)$$

Physical meaning

Inertia: the property that resists a change in velocity. The law holds in an **inertial frame**, such as the ground or a train at constant velocity.

Concept check: is the net force zero?

In which cases is $\sum \vec{F} = 0$?

(1) a football at the top of its flight
(2) a handcart gliding at constant speed on a frictionless tiled floor
(3) a passenger sitting in a train moving smoothly at constant velocity
(4) a car going round a bend at constant speed

Predict first

The first law concerns the *velocity vector*, not only the speed. In each case, does $\vec{v}$ change in size or direction? Which real objects exert forces?

Answer: (2) and (3).

(2), (3): $\vec{v}$ is constant, so $\vec{a} = 0$ and $\sum \vec{F} = 0$.
(1): $\vec{a} = \vec{g}$ at the top, so $\sum \vec{F} = m\vec{g} \neq 0$ (the Earth exerts the weight).
(4): the direction of $\vec{v}$ changes, so $\vec{a} \neq 0$ and $\sum \vec{F} \neq 0$.

Check: is the *vector* $\vec{v}$ constant?
But what if the speed is constant? A changing direction still needs a net force.

Decide first

A $70\,\mathrm{kg}$ passenger stands in a trotro moving at $15\,\mathrm{m/s}$. The trotro brakes hard at $4.0\,\mathrm{m/s^2}$ and the passenger lurches forward. Which force pushes the passenger forward?

Predict first

Draw the free-body diagram of the passenger. Which *outside* objects touch the passenger, and in which direction can they push?

Answer: none.

The only horizontal force is the friction of the floor on the feet, directed *backward*.
Its third-law partner is the feet on the floor, which acts on the floor, not on the passenger.
$f = ma \quad = 70 \times 4.0 \quad = 280\,\mathrm{N}$ backward.

Check: which object exerts a forward force? There is none.
The upper body tends to keep its earlier velocity (first law), so *relative to the trotro* it moves forward.

But what if the diagram needs a forward "inertial force"? It does not: the trotro is not an inertial frame; in the ground frame the lurch needs no forward force.

- **Mass** m (kg): a quantitative measure of inertia, that is, of how strongly an object resists acceleration. It is the same everywhere.
- **Weight** w (N): the total gravitational force on the object from all other bodies It depends on g; mass does not.

Rule: weight

$$w = mg \tag{3}$$

directed toward the Earth's centre;
$g = 9.8\,\mathrm{m/s^2}$ near the Earth's surface;
units $\mathrm{kg} \times \mathrm{m/s^2} = \mathrm{N}$.

Example

A $2.0\,\mathrm{kg}$ object: weight on Earth ($g = 9.8\,\mathrm{m/s^2}$) and on the Moon ($g = 1.62\,\mathrm{m/s^2}$)?

$w_{\text{Earth}} = 19.6\,\mathrm{N}$, $w_{\text{Moon}} = 3.2\,\mathrm{N}$; the mass is $2.0\,\mathrm{kg}$ in both places.

Free fall: $a = w/m = g$ for every mass, so a heavier object does not fall faster (Week 2).

Method: free-body diagram

1. **ISOLATE** one body and draw it as a dot.
2. **ASK** "Which other objects interact with this body?" Everything touching it, and the Earth.
3. **DRAW** one arrow per interaction, starting at the dot, in the direction the force acts *on the body*. Label each with its agent: "Earth on box (w)", "table on box (n)", and so on.
4. **AXES** are added afterwards (later in this contact, with the second law).

Rule

only forces acting *on* the body, each with a named agent.

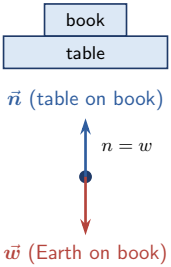
Watch out

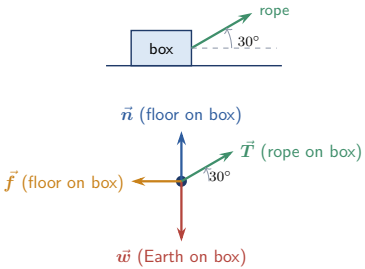
motion arrows are not forces: do not draw the net force, $m\vec{a}$ or the velocity, nor forces the body exerts on other objects.

Check: can I name an agent for every arrow, and have I missed any object touching the body?

Build it together: a book on a table, a box on a rope

CORE





Predict first

If the book is at rest, how do n and w compare? Why?

$\sum \vec{F} = 0$ (first law), so $n = w$: the table pushes up exactly as hard as the Earth pulls down.

Four interactions, four forces

Earth on box ($\vec{w}$), floor on box ($\vec{n}$ and $\vec{f}$), rope on box ($\vec{T}$). The rope's force is angled, so it will have components.

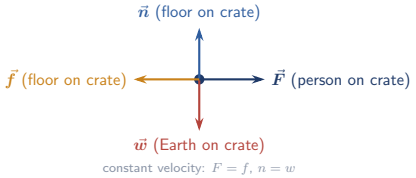
Your turn: draw the free-body diagrams

Independent problem

Draw a free-body diagram, with agent names, for each situation:
(a) a crate pushed along a rough floor at **constant velocity**;
(b) a block sliding down a **rough** incline.

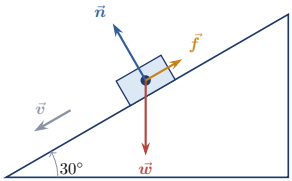
Predict first

For (a): how do the push and the friction compare? For (b): in which direction does friction point?



(a) A crate pushed at constant velocity

Four forces: $\vec{w}$ (Earth on crate), $\vec{n}$ (floor on crate), $\vec{F}$ (person on crate), $\vec{f}$ (floor on crate). At constant velocity $\sum \vec{F} = 0$, so the push equals the friction, and $n = w$.



(b) A block sliding down a rough incline

Three forces: $\vec{w}$, $\vec{n}$ (perpendicular to the surface) and $\vec{f}$ (up the slope, against the sliding).

Common error

There is no separate "force down the slope": that is a component of the weight, which we use in Contact B.

Rule: Newton's third law

If object A exerts a force on object B, then B exerts a force on A that is equal in magnitude and opposite in direction:

$$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}} \tag{6}$$

- The two forces are the **same type** and act at the same time: hand on wall and wall on hand; foot on football and football on foot.
- They act on **different bodies**, so they never appear in the same free-body diagram, and they never cancel.

Finding the partner of a force

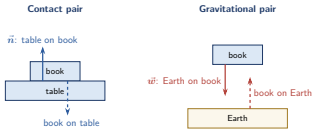
NAME it: "A on B". **FLIP** it: the partner is "B on A", the same type. **LOCATE** it: it acts on the *other* object. **COMPARE**: equal and opposite; the accelerations differ if the masses differ. Two forces of equal size are not a pair unless they are "A on B" and "B on A".

Predict first

A small car and a heavy truck collide head-on. Which experiences the larger force? Which has the larger acceleration? What is each reaction force: same object or different?

Concept check: pairs for the book on the table

CORE



Question

For the book at rest, are $\vec{w}$ (Earth on book) and $\vec{n}$ (table on book) a third-law pair?

Predict first

What is the partner of $\vec{n}$? Of $\vec{w}$? Does each partner act on the same object or a different one?

Answer: no. Both $\vec{w}$ and $\vec{n}$ act on the *book*.
Partner of $\vec{n}$ (table on book): the book on the table, acting on the *table*.
Partner of $\vec{w}$ (Earth on book): the book on the Earth, acting on the *Earth*.
Check: a pair is "A on B" and "B on A": two different objects.

But why are n and w equal here? Because $\sum \vec{F} = 0$ on the book (first law). In an accelerating lift $n \neq w$, but the pairs stay equal and opposite.

Question

A donkey pulls a cart forward. By the third law, the cart pulls back on the donkey with an equal force. **Why can the donkey and the cart accelerate forward?**

Predict first

Draw the free-body diagram of the cart, then of the donkey. Is the pair (donkey on cart, cart on donkey) on one diagram or on two?

Answer: the pair acts on different bodies.

Cart: forward pull of the donkey $>$ resistive forces, so the cart accelerates.
Donkey: forward force of the *ground* on its hooves $>$ backward pull of the cart.
The pair never appears twice on one diagram, so it never cancels.

Check: which object exerts each force on the body I am studying?

But what if a truck and a small car collide? The forces are equal; the car accelerates more because $a = F/m$.

Rule: equilibrium

An object at rest, or moving at constant velocity, has $\sum \vec{\boldsymbol{F}} = 0$:

$$\sum F_x = 0 \qquad \sum F_y = 0 \tag{5}$$

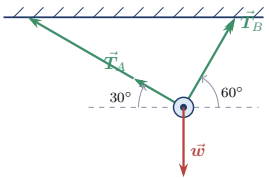
Equilibrium is the first law in components. It is also the second law, $\sum \vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}}$, with $\vec{\boldsymbol{a}} = 0$. *Why it matters:* it finds unknown forces (tensions, normal forces) in structures that do not move. Valid for rest or constant velocity; forces in N. Method: free-body diagram, axes, one equation per axis, right-hand side zero.

Predict first

A lamp hangs from two cables at different angles. Which cable carries more tension?

Question

A $10.0\,\mathrm{kg}$ lamp hangs at rest from two cables. Cable A makes 30° with the horizontal and cable B makes 60° on the other side. Find the tensions.

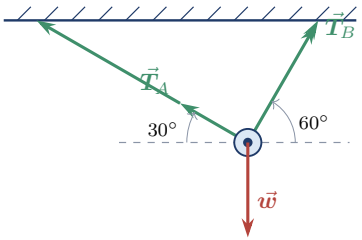


not to scale

Predict first

Which cable, A or B, carries more tension? Check this against the answer.

eg. cables supporting a load, as on the Golden Gate Bridge.



not to scale

Key idea

At rest the forces form a closed triangle: $\vec{T}_A + \vec{T}_B = -\vec{w}$.

IDENTIFY

Given: $m = 10.0\,\text{kg}$, cable A at $30°$, cable B at $60°$ to the horizontal, lamp at rest.

Find: T_A and T_B.

SET UP

Forces: $\vec{T}_A$, $\vec{T}_B$, $\vec{w}$ with $w = mg = 10.0(9.8) = 98.0\,\text{N}$ (3).

At rest, $\vec{a} = 0$: equilibrium (5).

$\vec{T}_A + \vec{T}_B + \vec{w} = 0 \iff \vec{T}_A + \vec{T}_B = -\vec{w}$

Axes: x right, y up. Components:

$$\begin{pmatrix} -T_A \cos 30° \\ T_A \sin 30° \end{pmatrix} + \begin{pmatrix} T_B \cos 60° \\ T_B \sin 60° \end{pmatrix} + \begin{pmatrix} 0 \\ -98.0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

EXECUTE

x: $-T_A \cos 30^\circ + T_B \cos 60^\circ = 0$

y: $T_A \sin 30^\circ + T_B \sin 60^\circ - 98.0 = 0$

From x: $T_A = \dfrac{\cos 60^\circ}{\cos 30^\circ} T_B = 0.577\, T_B$

Put into y: $(0.577)(0.500) T_B + 0.866\, T_B = 98.0\,\text{N}$

Combine: $1.155\, T_B = 98.0\,\text{N}$, so $T_B = 98.0/1.155$

$$\boxed{T_B = 84.9\,\text{N}, \qquad T_A = 0.577(84.9) = 49.0\,\text{N}}$$

EVALUATE

y check: $49.0 \sin 30^\circ + 84.9 \sin 60^\circ = 24.5 + 73.5 = 98.0\,\text{N}$. ✓

$T_B > T_A$ (steeper cable), as predicted.

Common mistakes and the correct habit

Mistake	Instead
Forces with no agent; $m\vec{a}$ or $\vec{v}$ drawn	Only forces on the body, each named
Thinking motion needs a force	Force causes acceleration, not velocity
Confusing mass with weight	$w = mg$ depends on place; mass does not
Adding force magnitudes	Resolve, combine, reconstruct
Drawing forces the body exerts on others	Draw only what acts *on* the body

Take away: forces are interactions and come in third-law pairs on different bodies; the free-body diagram comes first; $\sum \vec{F} = 0$ means equilibrium.

But what if... we can draw the forces, but not yet predict the motion? We need a law linking the **net force** to the **acceleration**.

From representing forces to predicting motion

We can draw the forces on a body. **How do they determine its acceleration?** (Second law, equilibrium, third law, connected objects.)

Poll

The same $20\,\mathrm{N}$ horizontal push is applied to a $40\,\mathrm{kg}$ box and to an $80\,\mathrm{kg}$ box on a smooth floor. How do the accelerations compare?

Predict first

Equal? Half? Double? And how would the acceleration change if the push were doubled? Sketch the free-body diagram.

The $80\,\mathrm{kg}$ box accelerates half as much. Doubling the force doubles the acceleration: $a \propto F$ and $a \propto 1/m$. For the $40\,\mathrm{kg}$ box: $\vec{w}$ and $\vec{n}$ are vertical and cancel, so $\sum F_x = 20 = 40\,a_x$ and $a_x = 0.50\,\mathrm{m/s^2}$, along the push (and $0.25\,\mathrm{m/s^2}$ for $80\,\mathrm{kg}$). ✓

But what if… the push is not horizontal, or several forces act? We need a law for the **net force**.

Introduce

A loaded truck and an empty handcart move at the same speed. Which is harder to stop? What about the same truck moving more slowly?

Predict first

Does the effect of a force on a moving body depend on its mass, its velocity, or both?

Define: the **linear momentum** of a body of mass m and velocity $\vec{v}$ is

$$\vec{p} = m\vec{v} \tag{4a}$$

m in kg, $\vec{v}$ in m/s, so $\vec{p}$ in $\mathrm{kg\,m/s}$ $(= \mathrm{N\,s})$.

Momentum is a vector: it points along $\vec{v}$, with components $p_x = mv_x$ and $p_y = mv_y$.

Explain. A $1500\,\mathrm{kg}$ car at $20\,\mathrm{m/s}$ east has $p = 1500(20) = 3.0 \times 10^4\,\mathrm{kg\,m/s}$ east. Reversing the velocity reverses the sign: $p = -3.0 \times 10^4\,\mathrm{kg\,m/s}$.

Check: both mass and velocity matter, so a heavy, fast body is hardest to stop.

Rule: Newton's second law, general form

The net force on a body equals the rate of change of its momentum, in an inertial frame:

$$\sum \vec{\boldsymbol{F}} = \frac{\mathrm{d}\vec{\boldsymbol{p}}}{\mathrm{d}t} = \frac{\mathrm{d}(m\vec{\boldsymbol{v}})}{\mathrm{d}t} \tag{4b}$$

Derivation of $\sum \vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}}$ for constant mass (one step per line):

1. Start: $\sum \vec{\boldsymbol{F}} = \mathrm{d}(m\vec{\boldsymbol{v}})/\mathrm{d}t$.
2. The mass is constant, so it leaves the derivative: $\sum \vec{\boldsymbol{F}} = m\,\mathrm{d}\vec{\boldsymbol{v}}/\mathrm{d}t$.
3. By definition $\mathrm{d}\vec{\boldsymbol{v}}/\mathrm{d}t = \vec{\boldsymbol{a}}$.
4. Therefore $\boxed{\sum \vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}}}$ (4).

Explain. Equation (4) is the special case of (4b) for constant mass; it fails when the mass changes (a truck losing sand, a rocket), where (4b) must be used.
Check: units, $(\mathrm{kg\,m/s})/\mathrm{s} = \mathrm{kg\,m/s^2} = \mathrm{N}$. ✓ Momentum and impulse return in Lecture 7.

Rule: Newton's second law

$$\sum \vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}} \qquad \Longleftrightarrow \qquad \sum F_x = ma_x, \quad \sum F_y = ma_y \tag{4}$$

The acceleration has the direction of the net force and is inversely proportional to the mass. *Why it matters:* it turns a list of forces into a prediction of motion. Symbols: $\sum \vec{\boldsymbol{F}}$ in N, m in kg, $\vec{\boldsymbol{a}}$ in $\mathrm{m/s^2}$.

- $\sum \vec{\boldsymbol{F}}$ means the net force: all forces *on this object*.
- Unit: $1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$. Forces cause *changes* of velocity, not velocity itself.
- Special case: $\vec{\boldsymbol{a}} = 0$ gives equilibrium (first law).

Model assumptions

Valid in an inertial frame, for an object treated as a particle with constant mass. For changing mass use $\sum \vec{\boldsymbol{F}} = \mathrm{d}\vec{\boldsymbol{p}}/\mathrm{d}t$ (4b).

Introduce. Equation (4), $\sum \vec{F} = m\vec{a}$, says three things about any body.

Predict first

A $20\,\mathrm{N}$ push moves a $40\,\mathrm{kg}$ box. What happens to a if the push doubles? If the mass doubles?

1. Direction. A net force changes the velocity *in the direction of the force*: $\vec{a}$ points along $\sum \vec{F}$. A positive change of v is a positive acceleration; a negative change is a deceleration (retardation).

2. Proportional to force. For fixed m: $a \propto \sum F$.

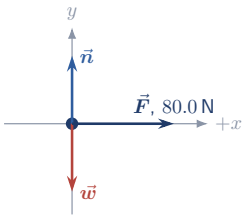
3. Inversely proportional to mass. For fixed force: $a \propto 1/m$.

Numbers (units $\mathrm{m/s^2}$):

$20\,\mathrm{N}$ on $40\,\mathrm{kg}$: $a = 20/40 = 0.50$
$40\,\mathrm{N}$ on $40\,\mathrm{kg}$: $a = 40/40 = 1.0$ (force doubled, a doubled)
$20\,\mathrm{N}$ on $80\,\mathrm{kg}$: $a = 20/80 = 0.25$ (mass doubled, a halved)

Example 3: a block of ice

CORE



smooth floor: no friction

Problem

A fish seller pushes a block of ice on a smooth floor with a constant horizontal force of $80.0\,\mathrm{N}$. Starting from rest, it moves $11.0\,\mathrm{m}$ in $5.00\,\mathrm{s}$.

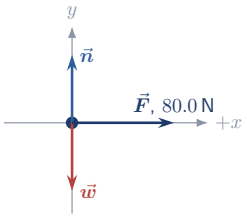
(a) What is the mass of the block?

(b) She stops pushing at $5.00\,\mathrm{s}$. How far does it slide in the next $5.00\,\mathrm{s}$?

Predict first

Predict: after she stops, does the block stop, slow down, or keep a constant velocity? Why?

eg. fish sellers at the Jamestown fishing harbour, Accra.



smooth floor: no friction

Key idea

Constant force, constant a.
When the push stops, $\sum F = 0$ and v stays constant.

IDENTIFY

Given: $F = 80.0\,\text{N}$, $v_{0x} = 0$, $x = 11.0\,\text{m}$ at $t = 5.00\,\text{s}$. Find: m, then the distance in the next $5.00\,\text{s}$.

SET UP

$+x$ along the push; y: $n = mg$. Smooth floor: $\sum F_x = F$, law (4): $F = ma_x$.

Constant a_x (Lecture 2): $x = v_{0x}t + \frac{1}{2}a_x t^2$.

EXECUTE

(a) $v_{0x} = 0$, so $x = \frac{1}{2}a_x t^2$ gives $a_x = \dfrac{2x}{t^2} = \dfrac{2(11.0\,\text{m})}{(5.00\,\text{s})^2} = 0.88\,\text{m/s}^2$

From $F = ma_x$: $m = \dfrac{F}{a_x} = \dfrac{80.0\,\text{N}}{0.88\,\text{m/s}^2}$ $\boxed{m = 90.9\,\text{kg}}$

(b) Speed when she stops: $v_x = a_x t = 0.88(5.00) = 4.4\,\text{m/s}$

No push: $\sum F_x = 0$ (2), so v_x stays $4.4\,\text{m/s}$

Distance: $x' = v_x t = 4.4(5.00)$ $\boxed{x' = 22\,\text{m}}$

EVALUATE

$x' = 2x$: constant speed after the push (first law). ✓

Decide first, and say which law you are using

(1) Can an object move while the net force on it is zero?
(2) Can the velocity be zero while the net force is not?
(3) Is the acceleration always in the direction of the velocity?
(4) If the net force doubles, what happens to a? If the mass doubles?

Predict first

Does a force set the *velocity* or the *change* of velocity? Which of $\vec{F}$, $\vec{v}$ and $\vec{a}$ always point the same way?

(1) **Yes.** Constant velocity: the first law.
(2) **Yes.** A football at the top of its flight: $\sum \vec{F} = m\vec{g}$.
(3) **No.** $\vec{a}$ points along the net force; a braking car has $\vec{a}$ opposite to $\vec{v}$.
(4) **It doubles; it halves.** $a = \sum F/m$.

Check: $\sum \vec{F} = m\vec{a}$ links force to *acceleration*, not to velocity.
But what if the object is already moving fast? Its velocity stays; only a net force changes it.

Method: axes, components, equations

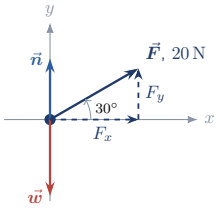
1. FBD. Draw the free-body diagram first (previous contact).
2. CHOOSE axes that simplify the physics, not automatically horizontal and vertical: one along the acceleration (or the motion), one perpendicular. On an incline, x is parallel to the surface and y perpendicular.
3. RESOLVE each force not along an axis into components (Week 1). Sketch the signs first: positive along the positive axis, negative against it.
4. WRITE one equation per axis, $\sum F_x = ma_x$ and $\sum F_y = ma_y$, *before* inserting numbers. Unknowns and equations must match; for equilibrium, $a = 0$.
5. SOLVE algebraically, then substitute with units. **6. EVALUATE:** sign, units, size, a limiting case.

WATCH OUT: a force with no component along an axis does not appear in that axis equation; n is found from the equation perpendicular to the surface, so do not assume $n = mg$.

CHECK: do the components, recombined, give back the original force? Does the direction of $\vec{a}$ agree with the direction of the net force in the diagram?

Question

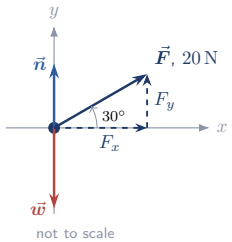
The same $40\,\mathrm{kg}$ box is now pulled by a rope with a $20\,\mathrm{N}$ force at 30° above the horizontal. The floor is still smooth. Find the acceleration and the normal force.



Predict first

Will a be larger or smaller than $0.50\,\mathrm{m/s^2}$? Will n be larger or smaller than mg?

Example 4: a pulled box, solution



Key idea

Only the horizontal part of the pull accelerates the box; the vertical part reduces n.

IDENTIFY

Given: $m = 40\,\text{kg}$, $F = 20\,\text{N}$ at $30°$, smooth floor. Find: a_x and n.

SET UP

Forces: $\vec{n}$, $\vec{w}$, $\vec{F}$. Law (4), the box accelerates: $\vec{n} + \vec{w} + \vec{F} = m\vec{a}$

Axes: x along the motion, y up; $F_x = F\cos 30°$, $F_y = F\sin 30°$.

EXECUTE

x: $F\cos 30° = ma_x$, so

$$a_x = \frac{F\cos 30°}{m} = \frac{(20\,\text{N})(0.866)}{40\,\text{kg}} = \frac{17.3\,\text{N}}{40\,\text{kg}}$$

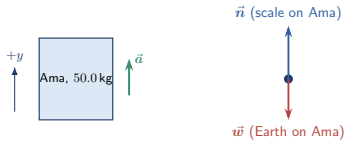
y: $n + F\sin 30° - mg = 0$, so

$$n = mg - F\sin 30° = 392\,\text{N} - 10.0\,\text{N}$$

$$\boxed{a_x = 0.43\,\text{m/s}^2, \quad n = 382\,\text{N}}$$

EVALUATE

$a < 0.50\,\text{m/s}^2$ and $n < mg = 392\,\text{N}$, as predicted. ✓



Problem

A lift and its load have a total mass of $800\,\mathrm{kg}$. It moves down at $10.0\,\mathrm{m/s}$ and stops with constant acceleration over $25.0\,\mathrm{m}$.

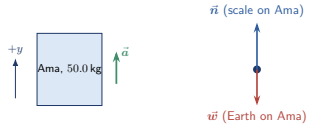
(a) What is the tension in the supporting cable?

(b) Ama, of mass $50.0\,\mathrm{kg}$, stands on a bathroom scale in the lift. What does the scale read (her apparent weight) while it stops?

Predict first

Predict: is the cable tension larger or smaller than $mg = 7840\,\mathrm{N}$? Does the scale read more or less than mg?

eg. the lifts of the Burj Khalifa, Dubai.



Key idea

The scale reads n, not mg: $n = m(g + a_y)$.

IDENTIFY

Given: $m = 800\,\text{kg}$, $v_{0y} = -10.0\,\text{m/s}$, $v_y = 0$, $y - y_0 = -25.0\,\text{m}$ ($+y$ up). Find: T; the scale reading n for $m_A = 50.0\,\text{kg}$.

SET UP

Kinematics (Lecture 2): $v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$. Law (4): $\sum F_y = ma_y$.

Lift: $\vec{\boldsymbol{T}} + \vec{\boldsymbol{w}} = m\vec{\boldsymbol{a}}$ Ama: $\vec{\boldsymbol{n}} + \vec{\boldsymbol{w}}_A = m_A\vec{\boldsymbol{a}}$

EXECUTE

Solve for a_y: $a_y = \dfrac{v_y^2 - v_{0y}^2}{2(y - y_0)} = \dfrac{0 - (-10.0\,\text{m/s})^2}{2(-25.0\,\text{m})} = +2.00\,\text{m/s}^2$

(a) Lift: $T - mg = ma_y$, so $T = m(g + a_y) = 800(9.8 + 2.00)\,\text{N}$

$\boxed{T = 9.44 \times 10^3\,\text{N}}$

(b) Ama: $n - m_A g = m_A a_y$, so

$n = m_A(g + a_y) = 50.0(9.8 + 2.00)\,\text{N}$ $\boxed{n = 590\,\text{N}}$

EVALUATE

$T > mg = 7840\,\text{N}$ and $n > m_A g = 490\,\text{N}$, as predicted. ✓

Common mistakes in Contact A (the second law)

Common mistakes and the correct habit

Mistake	Instead
Assuming $n = mg$ everywhere	Get n from the perpendicular equation
Putting a third-law pair on one body	The pair acts on different bodies; no cancelling
Always using horizontal and vertical axes	Put an axis along $\vec{a}$ or the slope
A force perpendicular to an axis	It has no component along that axis

Take away: net force causes acceleration; mass resists it; momentum is $\vec{p} = m\vec{v}$ and $\sum \vec{F} = \mathrm{d}\vec{p}/\mathrm{d}t$.

Contact A: what you can now do

- Name every force by its agent and draw a free-body diagram holding only the forces on the body.
- Find a net force by components; use the first law, equilibrium and third-law pairs.
- Write $\sum \vec{F} = m\vec{a}$ by components for one body, get n from the perpendicular equation, and read apparent weight in a lift.

Predict first

Before Wednesday, predict: a rope joins two blocks on a smooth floor and you pull one of them. Do both blocks have the same acceleration? Is the tension in the rope equal to your pull? And what does a rough floor do to a heavy crate you push gently?

But what if... several bodies are joined by ropes, and the floor is rough? → **connected objects and friction**.

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Contact B: Analyse, then Apply

How do the laws apply to connected bodies and friction, and then to slopes, curves and combined systems?

Predict first

1. A $5.0\,\mathrm{kg}$ box on a smooth floor has $20\,\mathrm{N}$ pulling east and $5.0\,\mathrm{N}$ pulling west. Find $\vec{\boldsymbol{a}}$. $\sum F_x = 20 - 5.0 = 15\,\mathrm{N}$, so $a_x = 15/5.0 = 3.0\,\mathrm{m/s^2}$ east (second law).
2. A book rests on a table. Are $\vec{\boldsymbol{w}}$ and $\vec{\boldsymbol{n}}$ a third-law pair? No: both act on the book. The partner of $\vec{\boldsymbol{n}}$ is the book on the table.
3. A rope pulls a $40\,\mathrm{kg}$ box at 30° above the horizontal. Is n larger or smaller than mg? Smaller: the vertical part of the pull supports part of the weight, $n = mg - F\sin 30^\circ$.

Rule: ideal rope and pulley

An **ideal rope** is light and does not stretch: it has the same tension throughout, and objects joined by a taut rope have the same acceleration magnitude. An **ideal pulley** is light and frictionless and only changes the direction of the rope.

(7) $|a_1| = |a_2|$, T the same throughout the rope (a in $\mathrm{m/s^2}$, T in N).

Physical meaning

Real ropes have mass and stretch, and real pulleys have friction; the ideal model is valid when these are small. A rope can pull but not push, and T is not automatically a weight: it depends on the acceleration.

Predict first

Predict: a taut rope joins two blocks and one is pulled. If the rope were cut, what happens to the other block's acceleration? What does that say about T?

Method: connected objects

1. **ISOLATE** each body and draw its own free-body diagram.
2. **CONSTRAINT:** state the link: same acceleration magnitude, same rope tension (ideal rope and pulley).
3. **AXES:** choose each body's axis along its own motion, with signs consistent with the constraint.
4. **WRITE** one $\sum F = ma$ equation per body along its motion.
5. **SOLVE** the simultaneous equations for a and T.

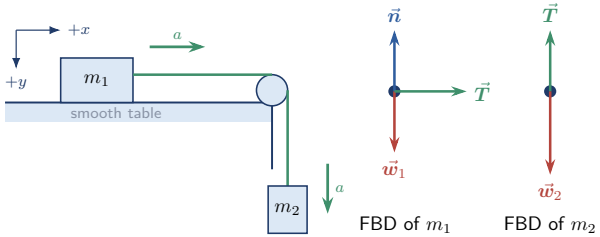
Watch out

drawing one diagram for the whole system and forgetting that internal tensions appear only when you separate the bodies.

Check

treat the whole system once: the external force equals the total mass times a. Check limiting cases.

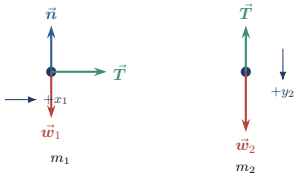
A block $m_1 = 3.0\,\mathrm{kg}$ on a smooth table is joined by an ideal rope over an ideal pulley to a hanging block $m_2 = 2.0\,\mathrm{kg}$. Released from rest, find the acceleration and the tension.



Predict first

Is T larger or smaller than $m_2 g = 19.6\,\mathrm{N}$? Is a larger or smaller than g?

eg. the Atwood machine used in school physics laboratories.



Key idea

Same a and same T for both blocks; $T < m_2 g$ because m_2 accelerates.

IDENTIFY

Given: $m_1 = 3.0\,\text{kg}$ (smooth table), $m_2 = 2.0\,\text{kg}$ hanging, ideal rope and pulley. Find: a and T.

SET UP

Law (4) for each block; ideal rope (7): same a, same T.

m_1: $\vec{n} + \vec{w}_1 + \vec{T} = m_1\vec{a}$ $\qquad$ m_2: $\vec{T} + \vec{w}_2 = m_2\vec{a}$

Axes: $+x_1$ toward the pulley; $+y_2$ downward.

EXECUTE

m_1, x_1: $T = m_1 a$ $\qquad$ m_1, y: $n - m_1 g = 0$

m_2, y_2: $m_2 g - T = m_2 a$

Add the two (the T cancels): $m_2 g = (m_1 + m_2)a$

Solve: $a = \dfrac{m_2 g}{m_1 + m_2} = \dfrac{2.0(9.8)\,\text{N}}{5.0\,\text{kg}} = 3.92\,\text{m/s}^2$

Back-substitute: $T = m_1 a = 3.0(3.92)\,\text{N}$

$$\boxed{a = 3.9\,\text{m/s}^2, \quad T = 11.8\,\text{N}}$$

EVALUATE

$T < m_2 g = 19.6\,\text{N}$ and $a < g$, as predicted. ✓

From smooth floors to real surfaces

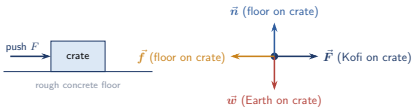
So far every floor was smooth. **What does a real, rough surface do to a sliding or resting body?**

Poll

You push a heavy crate. Nothing happens. You push harder: still nothing. Harder again, and it suddenly slides. How does the floor's friction force change as you push harder?

Predict first

Sketch friction against applied force before the next slide. If friction disappeared right now, what would happen to the crate, and to you pushing it?



The situation. Kofi pushes the crate harder and harder along a rough concrete floor.

The question. How does the floor's friction force respond to the push?

What the graph will show. Friction f (vertical axis, N) against applied force F (horizontal axis, N).

Friction force

Define: the contact force exerted by a surface on an object, *parallel* to the surface, opposing the relative sliding (or the tendency to slide). Unit: N. The **normal force** n is the perpendicular contact force of the same surface. Two types of friction follow.



(a) no push (b) weak push (c) about to slide (d) sliding

Predict first

Predict: as Kofi pushes harder from (a) to (d), what happens to the friction on the crate?

(a) No push: $F = 0$, so $f = 0$. Friction does not exist until something tries to slide the surfaces.
(b) Weak push: the crate stays at rest, $\sum F_x = 0$, so $f_s = F$: static friction grows with the push.
(c) About to slide: f_s has reached its largest value, $f_{s,\max} = \mu_s n$.
(d) Sliding: the friction drops to $f_k = \mu_k n$, smaller than $\mu_s n$; the net force is now $F - f_k$.

Static friction

Definition:
friction when the two surfaces are not sliding; it adjusts to match the other forces.

Equation:
$0 \leq f_s \leq \mu_s n$ (8a)

Symbols and units:
f_s in N; n in N; μ_s the coefficient of static friction, no units. $f_s = \mu_s n$ only at the point of sliding.

Kinetic friction

Definition:
friction when the two surfaces slide over each other; it equals $\mu_k n$ and does not depend on the applied force.

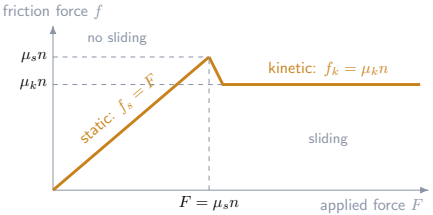
Equation:
$f_k = \mu_k n$ (8b)

Symbols and units:
f_k in N; n in N; μ_k the coefficient of kinetic friction, no units. Usually $\mu_k < \mu_s$.

Both act *parallel* to the surface, against the relative sliding (or the tendency to slide). The normal force n comes from the equation perpendicular to the surface, not automatically mg.

Friction (4): explain the graph

CORE



Predict first

What does f do as F grows from zero, before the crate moves?

1. Rising line: the crate is at rest, so $\sum F_x = 0$ *Reuse:* (5) and $f_s = F$: friction matches the push.

2. Peak: at $F = \mu_s n$ static friction reaches its maximum (8a); the crate is about to slide.

3. Drop: once it slides, friction falls to $f_k = \mu_k n$ (8b), smaller since $\mu_k < \mu_s$.

4. Flat line: f_k does not depend on F while the crate slides.

$f_s = \mu_s n$ is the maximum static friction, not its value

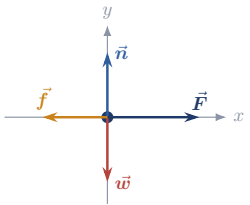
Static friction is self-adjusting: $0 \leq f_s \leq \mu_s n$. It equals $\mu_s n$ only at the threshold of motion. Once the object slides, friction becomes kinetic: $f_k = \mu_k n$.

Question

A crate of bottled drinks weighing $500\,\mathrm{N}$ rests on a level concrete floor. Kofi must pull horizontally with $230\,\mathrm{N}$ to start it moving, and then $200\,\mathrm{N}$ keeps it moving at constant velocity. (a) Find μ_s and μ_k. (b) If he pulls with only $50\,\mathrm{N}$, what is the friction? (c) He now pulls on a rope 30° above the horizontal. How hard must he pull to keep it moving at constant velocity?

Predict first

Sketch the FBD. Which is larger, μ_s or μ_k? Will the upward-angled pull need more or less than $200\,\mathrm{N}$?



Key idea

Static friction matches the pull up to $\mu_s n$; sliding friction is $\mu_k n$.

IDENTIFY

Given: $w = 500\,\text{N}$; $F = 230\,\text{N}$ just starts the crate; $200\,\text{N}$ keeps it moving. Find: μ_s, μ_k, f_s at $50\,\text{N}$, and T at $30°$.

SET UP

At rest or at constant velocity $a = 0$: $\vec{n} + \vec{w} + \vec{F} + \vec{f} = \vec{0}$ (4), (8a), (8b)

Axes: $+x$ along the pull, y up. y: $n - w = 0$ *Reuse:* (4) with $a_y = 0$.

EXECUTE

$n = w = 500\,\text{N}$

(a) Just before sliding $f_s = (f_s)_{\max} = 230\,\text{N}$, so $\mu_s = 230/500 = \boxed{0.46}$; moving, $f_k = 200\,\text{N}$, so $\mu_k = 200/500 = \boxed{0.40}$

(b) $50 < 230$: no sliding, $a_x = 0$, so $\boxed{f_s = F = 50\,\text{N}}$

(c) x: $T\cos 30° - \mu_k n = 0$; y: $n + T\sin 30° - w = 0$

$$T = \frac{\mu_k w}{\cos 30° + \mu_k \sin 30°} = \frac{0.40(500)}{0.866 + 0.200} = \boxed{188\,\text{N}},$$

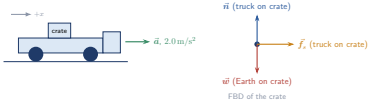
$n = 500 - 188(0.500) = 406\,\text{N}$

EVALUATE

(a) $\mu_k < \mu_s$ ✓ (c) The pull lifts the crate a little, so n and f_k fall and $188\,\text{N} < 200\,\text{N}$. ✓

Question

A crate of tomatoes sits on the bed of a pickup truck that accelerates forward at $2.0\,\mathrm{m/s^2}$. The crate does not slide. What force accelerates the crate? What is the least μ_s?



Predict first

Which *outside* object touches the crate, and which way does it push?

Answer: static friction of the truck on the crate, forward.

Its partner is the crate on the truck, acting on the truck.

$f_s = ma$ *Reuse:* (4), x-component $= m(2.0)$.

$f_s \le \mu_s n = \mu_s mg$ *Reuse:* (8a), (3), so $\mu_s \ge a/g = 2.0/9.8 = 0.20$.

Check: friction opposes *sliding* between surfaces; it can point along the motion.

But what if the truck brakes? Friction then points backward on the crate.

Think before calculating

A car accelerates from rest on a level road ($\mu_s = 0.80$). Which force on the car accelerates it?
(a) the force of the engine
(b) the friction of the road on the tyres
(c) the force of the tyres on the road
(d) the weight of the car

Predict first

Can an object accelerate itself using only forces between its own parts? Which object *outside* the car pushes it forward?

Answer: (b). The engine turns the driven wheels; the tyres push *backward* on the road; by the third law the road pushes *forward* on the tyres. This is **static friction**, and it acts on the car.

Check: which object exerts the force? In (c) it is the tyres, and it acts *on the road*, so it cannot accelerate the car.

Limit: $f_s \le \mu_s n$ gives $a_{\max} = \mu_s n/m = \mu_s g = 0.80(9.8) \approx 7.8\,\mathrm{m/s^2}$.
But what if the road is wet and muddy? μ_s falls, little friction is available, and a powerful engine cannot help.

From level floors to slopes and curves

The laws do not change; the forces that appear do. **How do we apply Newton's laws on a tilted surface, on a curved path, and in a system with several effects at once?**

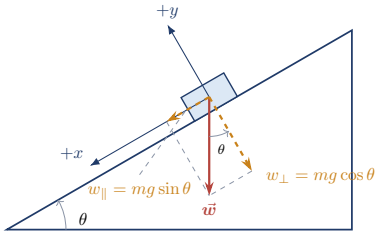
This contact: **inclined planes**; **circular dynamics** (flat, banked, conical and vertical); one integrated problem.

Poll

A car goes round a bend at constant speed. Is its velocity constant? Is the net force on it zero?

Predict first

A block slides down a smooth slope. Which direction does the net force point, and how does the acceleration compare with g?



INTRODUCE. On a slope the weight points neither along nor across the surface, so we resolve it on axes chosen *on the slope*: the motion is then along one axis.

Rule: weight on an incline

DEFINE. θ: angle of the slope above the horizontal (degrees); x along the slope, y perpendicular; $w = mg$.

$$w_\parallel = mg\sin\theta, \quad w_\perp = mg\cos\theta \tag{9}$$

Predict first

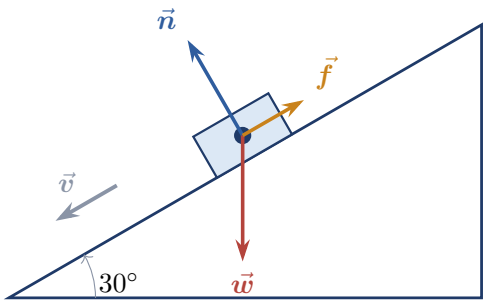
EXPLAIN. Predict: on a steeper slope, which component grows? Limits: $\theta \to 0$, $w_\parallel \to 0$; $\theta \to 90°$, $w_\perp \to 0$. Example: $5.0\,\mathrm{kg}$, $30°$: $24.5\,\mathrm{N}$, $42.4\,\mathrm{N}$.

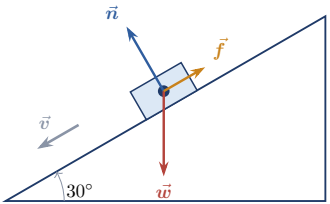
Question

A $5.0\,\mathrm{kg}$ block slides down a 30.0° incline with kinetic friction $\mu_k = 0.30$. Find its acceleration.

Predict first

Is a larger or smaller than $g \sin 30^\circ = 4.9\,\mathrm{m/s^2}$? Does the answer depend on the mass?





Key idea

Along the slope $mg\sin\theta$ drives and f_k opposes; across it $n = mg\cos\theta$.

IDENTIFY

Given: $m = 5.0\,\text{kg}$, $\theta = 30.0°$, $\mu_k = 0.30$. Find: a.

SET UP

Forces: $\vec{\boldsymbol{w}}$, $\vec{\boldsymbol{n}}$ (perpendicular), $\vec{\boldsymbol{f}}_k$ (up the slope). Law (4): $\vec{\boldsymbol{n}} + \vec{\boldsymbol{w}} + \vec{\boldsymbol{f}}_k = m\vec{\boldsymbol{a}}$

Axes: x down the slope, y perpendicular. (9): $w_\parallel = mg\sin\theta$, $w_\perp = mg\cos\theta$ *Reuse:* components as in (1); (3).

EXECUTE

y: $n - mg\cos\theta = 0$, so $n = 5.0(9.8)\cos 30° = 42.4\,\text{N}$

$f_k = \mu_k n = 0.30(42.4) = 12.7\,\text{N}$ (8b)

Weight along the slope: $mg\sin\theta = 5.0(9.8)\sin 30° = 24.5\,\text{N}$

x: $mg\sin\theta - f_k = ma$, so $a = \dfrac{24.5\,\text{N} - 12.7\,\text{N}}{5.0\,\text{kg}}$

$\boxed{a = 2.4\,\text{m/s}^2 \text{ down the slope}}$

EVALUATE

$a < g\sin 30° = 4.9\,\text{m/s}^2$, as predicted; $a = g(\sin\theta - \mu_k\cos\theta)$ is independent of m. ✓

Poll

A car goes round a flat circular bend at a **constant speed**. Is the net force on it zero? If not, in which direction does it point?

Predict first

Recall from Week 2: what are the direction and size of the acceleration in uniform circular motion? If the string on a whirled stone snapped, which way would the stone go, and what force would remain?

Answer

The acceleration is v^2/R toward the centre, so the net force points to the centre. If the string snapped, no net force would remain: by the first law the stone moves in a straight line at constant velocity, along the tangent to the circle.

But what if... the acceleration v^2/R toward the centre is known, but *which force* supplies it? We need the dynamics of circular motion.

Rule: net force in uniform circular motion

In uniform circular motion (constant v, radius R) the velocity keeps changing direction, so $a_{\text{rad}} = v^2/R$ toward the centre (Week 2). The net force must point toward the centre *Reuse:* (4), radial axis; m (kg), v (m/s), R (m), ω (rad/s):

$$\sum F_{\text{radial}} = m\,\frac{v^2}{R} = m\omega^2 R \tag{10}$$

Friction supplies the centripetal force

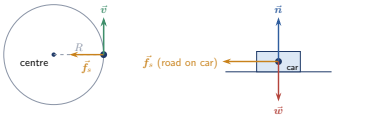
"Centripetal force" is whichever real force supplies the inward net force: gravity (satellite), tension (stone on a string), friction (car on a flat bend).
On a flat curve static friction is the only horizontal force, so $f_s = mv^2/R$.
The car stays on only if $f_s \le \mu_s n$; this fixes the least μ_s or the greatest safe speed.

Method: circular dynamics

1. FBD, real forces only. **2.** Radial axis toward the centre. **3.** Write $\sum F_{\text{radial}} = mv^2/R$ and the perpendicular equation. **4.** Name the real force supplying it. **5.** Apply its limit to find $\mu_{s,\min}$ or $v_{\max}$.

Question

A $1200\,\mathrm{kg}$ car rounds a flat circular bend (on a village road) of radius $50.0\,\mathrm{m}$ at $15.0\,\mathrm{m/s}$. (a) What friction force is required, and what is the least μ_s? (b) What is the maximum speed for $\mu_s = 0.80$ (dry) and $\mu_s = 0.30$ (wet)?



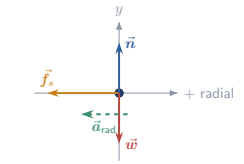
Predict first

Predict:

1. What force can point toward the centre on a flat road?
2. If the car goes faster, is more or less friction required?
3. Is there a speed above which the car must skid?

eg. bends along the Accra–Kumasi road.

Example 9: solution (1): identify and set up



Key idea

Static friction supplies the centripetal force, $f_s = mv^2/R$. It cannot exceed $\mu_s n$: at the limit $mv^2/R = \mu_s n$, which gives the least μ_s or the greatest v.

IDENTIFY

Given: $m = 1200\,\text{kg}$, $R = 50.0\,\text{m}$, $v = 15.0\,\text{m/s}$
$\mu_s = 0.80$ (dry), $\mu_s = 0.30$ (wet)
Find: f_s, $\mu_{s,\text{min}}$, v_{max}

SET UP

$\vec{n} + \vec{w} + \vec{f}_s = m\vec{a}$

radial: $f_s = \dfrac{mv^2}{R}$ (4), (10)

limit: $f_s \leq \mu_s n$ (8a)

y: $n = mg = 1200(9.8) = 11\,760\,\text{N}$

EXECUTE

(a) $f_s = \dfrac{mv^2}{R} = \dfrac{1200(15.0)^2}{50.0} = 5400\,\text{N}$

limit $\mu_s n = f_s$: $\quad \mu_{s,\min} = \dfrac{5400}{11\,760} = 0.46$

(b) limit $\mu_s mg = \dfrac{mv_{\max}^2}{R}$: $\quad v_{\max} = \sqrt{\mu_s g R}$

dry: $\sqrt{0.80(9.8)(50.0)} = 19.8\,\text{m/s}$

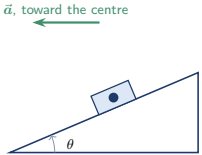
wet: $\sqrt{0.30(9.8)(50.0)} = 12.1\,\text{m/s}$

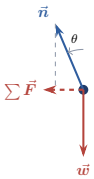
$\boxed{f_s = 5400\,\text{N}}$ $\quad \boxed{\mu_{s,\min} = 0.46}$ $\quad \boxed{v_{\max} = 19.8\,\text{m/s (dry)},\ 12.1\,\text{m/s (wet)}}$

EVALUATE

$15.0 < 19.8$ (dry) but $15.0 > 12.1$ (wet): safe when dry, skids when wet. ✓

Example 10: a banked bend





Introduce

A motorway bend of radius $150\,\mathrm{m}$ is banked at an angle θ so that cars can take it at $25.0\,\mathrm{m/s}$ with no friction at all. Find θ.

Define: the **bank angle** θ is the angle of the road surface above the horizontal; $\vec{n}$ is perpendicular to the surface, so it tilts toward the centre.

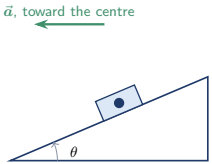
Why: with no friction, the horizontal component of $\vec{n}$ is the only inward force, so it supplies the centripetal force.

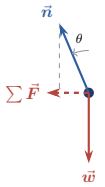
eg. Aburi Mountain Road (N4); Daytona Speedway (31° banking).

Predict first

Without friction, which force supplies the inward net force? What happens to the needed θ if v doubles?

Example 10: a banked bend, solution





Key idea

No friction: the horizontal part of n supplies the centripetal force, $n \sin\theta = mv^2/R$.

IDENTIFY

Given: $R = 150\,\text{m}$, $v = 25.0\,\text{m/s}$, frictionless. Find: θ.

SET UP

Forces: $\vec{n}$ (perpendicular to the road), $\vec{w}$. Law (4) with (10): $\vec{n} + \vec{w} = m\vec{a}$.

Axes: x horizontal toward the centre, y vertical.

EXECUTE

x: $n \sin\theta = mv^2/R$

y: $n \cos\theta - mg = 0$

Divide x by y (n cancels): $\dfrac{n \sin\theta}{n \cos\theta} = \dfrac{mv^2/R}{mg}$, so $\tan\theta = \dfrac{v^2}{gR}$

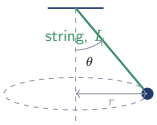
Substitute: $\tan\theta = \dfrac{(25.0\,\text{m/s})^2}{(9.8\,\text{m/s}^2)(150\,\text{m})} = 0.425$

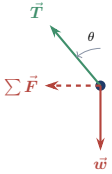
$$\boxed{\theta = \tan^{-1}(0.425) = 23.0^\circ}$$

EVALUATE

m cancels; $v = 0$ gives $\theta = 0$; $\tan\theta \propto v^2$. ✓

Example 11: a conical pendulum





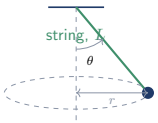
Introduce

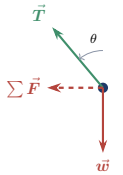
Yaw whirls a stone of mass $0.40\,\mathrm{kg}$ on a light string of length $L = 1.50\,\mathrm{m}$ in a *horizontal* circle. The string makes $\theta = 40.0^\circ$ with the vertical. Find T and ω.

Define: a **conical pendulum** moves in a horizontal circle of radius $r = L\sin\theta$; the string sweeps out a cone.
Why it works: the horizontal component of $\vec{T}$ is the only inward force; the vertical component balances the weight.
eg. a fairground chair-swing ride.

Predict first

The stone does not move vertically. What does that say about the vertical components of the forces? Is T larger or smaller than mg?





Key idea

The horizontal part of T supplies the centripetal force; the vertical part balances w.

IDENTIFY

Given: $m = 0.40\,\text{kg}$, $L = 1.50\,\text{m}$, $\theta = 40.0°$. Find: T and ω.

SET UP

Forces: $\vec{\boldsymbol{T}}$ (along the string), $\vec{\boldsymbol{w}}$ (downwards).

$\vec{\boldsymbol{T}} + \vec{\boldsymbol{w}} = m\vec{\boldsymbol{a}}$ (4),where $a_{\text{rad}} = \omega^2 r$ (10), $r = L\sin\theta$.

Axes: x toward the centre (horizontal), y up.

EXECUTE

y: $T\cos\theta - mg = 0$, so $T = \dfrac{0.40(9.8)}{\cos 40.0°} = \dfrac{3.92}{0.766}$

x: $T\sin\theta = m\omega^2 r = m\omega^2 L\sin\theta$; cancel $\sin\theta$: $T = m\omega^2 L$

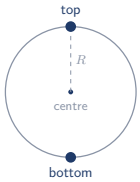
Solve: $\omega^2 = \dfrac{T}{mL} = \dfrac{5.12\,\text{N}}{(0.40\,\text{kg})(1.50\,\text{m})} = 8.53\,\text{rad}^2/\text{s}^2$

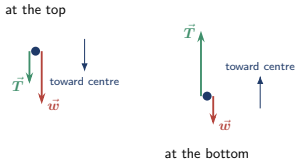
$$\boxed{T = 5.1\,\text{N}, \qquad \omega = 2.9\,\text{rad/s}}$$

EVALUATE

$T > mg = 3.92\,\text{N}$, as predicted. ✓ Check: $\omega = \sqrt{g/(L\cos\theta)} = 2.92\,\text{rad/s}$, period $2\pi/\omega = 2.15\,\text{s}$.

Example 12: a vertical circle





Introduce

Daavi whirls a stone of mass $0.30\,\mathrm{kg}$ on a string in a *vertical* circle of radius $R = 1.20\,\mathrm{m}$. Its speed is $4.5\,\mathrm{m/s}$ at the top and $7.0\,\mathrm{m/s}$ at the bottom. Find the tension at each point, and the least speed at the top for the string to stay taut.

Predict first

At which point, top or bottom, is the string more likely to go slack? Which forces point toward the centre at each point?

eg. whirling a bucket of water in a vertical circle; the loop of a roller coaster.







Key idea

T and w combine toward the centre: $T + mg$ at the top, $T - mg$ at the bottom, $= mv^2/R$.

IDENTIFY

Given: $m = 0.30\,\text{kg}$, $R = 1.20\,\text{m}$, $v_t = 4.5$, $v_b = 7.0\,\text{m/s}$. Find: T_t, T_b, $v_{\min}$.

SET UP

Law (4) with (10); the axis points toward the centre.

Top: $\vec{\boldsymbol{T}}$ and $\vec{\boldsymbol{w}}$ both point toward the centre: $T + mg = mv^2/R$.

Bottom: $\vec{\boldsymbol{T}}$ toward the centre, $\vec{\boldsymbol{w}}$ away: $T - mg = mv^2/R$.

EXECUTE

top: $T_t = m\left(\dfrac{v_t^2}{R} - g\right) = 0.30\left(\dfrac{4.5^2}{1.20} - 9.8\right)\ \text{N}$ $\boxed{T_t = 2.12\,\text{N}}$

bottom: $T_b = m\left(\dfrac{v_b^2}{R} + g\right) = 0.30\left(\dfrac{7.0^2}{1.20} + 9.8\right)\ \text{N}$

$\boxed{T_b = 15.2\,\text{N}}$

taut: $T_t \geq 0$ gives $v_t^2/R \geq g$, so $v_{\min} = \sqrt{gR} = \sqrt{9.8(1.20)}$

$\boxed{v_{\min} = 3.43\,\text{m/s}}$

EVALUATE

$T_b > T_t$: the string slackens first at the top, as predicted. ✓

In the car, you feel thrown outward.

Is there an outward force pushing you?

Predict first

Which *outside* object touches you? In which direction can the seat and door push?

Answer: no outward force acts on you.

In the ground (inertial) frame your body tends to move in a straight line (first law).
The car turns toward the centre, so the door pushes you *inward*.
That inward net force makes you follow the curve; the “outward push” is your inertia.

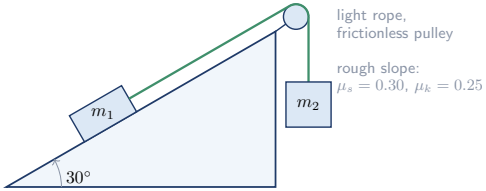
Check: which object exerts the outward force? There is none.

But what if the door were not there? You would continue in a straight line, tangent to the bend.

Application problem: decide what physics applies

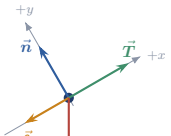
A $10.0\,\mathrm{kg}$ crate rests on a rough 30° incline ($\mu_s = 0.30$, $\mu_k = 0.25$). A light rope runs from the crate, parallel to the slope, over a frictionless pulley to an $8.0\,\mathrm{kg}$ hanging block. The system is released from rest.

Does the system move? If so, in which direction? Find the acceleration and the tension.

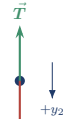


Predict first

IDENTIFY → FBDs and axes → PREDICT → EXECUTE → EVALUATE.



crate m_1



block m_2

Key idea

A slope, friction and a rope: one second-law equation per body, same a and T.

IDENTIFY

Given: $m_1 = 10.0\,\text{kg}$ on a 30° slope, $\mu_s = 0.30$, $\mu_k = 0.25$; $m_2 = 8.0\,\text{kg}$ hanging; released from rest.

Find: does it move? a and T.

SET UP

Law (4) for each body; constraint (7); friction (8); incline components (9).

crate: $\vec{\boldsymbol{n}} + \vec{\boldsymbol{w}}_1 + \vec{\boldsymbol{T}} + \vec{\boldsymbol{f}} = m_1\vec{\boldsymbol{a}}$

block: $\vec{\boldsymbol{T}} + \vec{\boldsymbol{w}}_2 = m_2\vec{\boldsymbol{a}}$

Axes: crate $+x$ up the slope, $+y$ perpendicular; block $+y_2$ down.

EXECUTE: does it move?

$n = m_1 g \cos 30^\circ = 10.0(9.8)\cos 30^\circ = 84.9\,\text{N}$
Net pull toward the block: $m_2 g - m_1 g \sin 30^\circ = 78.4 - 49.0 = 29.4\,\text{N}$
Largest static friction (8a): $\mu_s n = 0.30(84.9) = 25.5\,\text{N}$
$29.4 > 25.5$: it moves, the crate going **up** the slope.

EXECUTE: acceleration and tension

crate, x: $T - m_1 g \sin 30^\circ - f_k = m_1 a$ block: $m_2 g - T = m_2 a$ $f_k = \mu_k n$
$f_k = 0.25(84.9) = 21.2\,\text{N}$
Add the two equations (T cancels): $m_2 g - m_1 g \sin 30^\circ - f_k = (m_1 + m_2)a$
Solve: $a = \dfrac{78.4 - 49.0 - 21.2}{18.0}\,\text{m/s}^2 = \dfrac{8.18}{18.0}\,\text{m/s}^2$
From the block: $T = m_2(g - a) = 8.0(9.8 - 0.455)\,\text{N}$

$$\boxed{a = 0.45\,\text{m/s}^2, \quad T = 74.8\,\text{N}}$$

EVALUATE

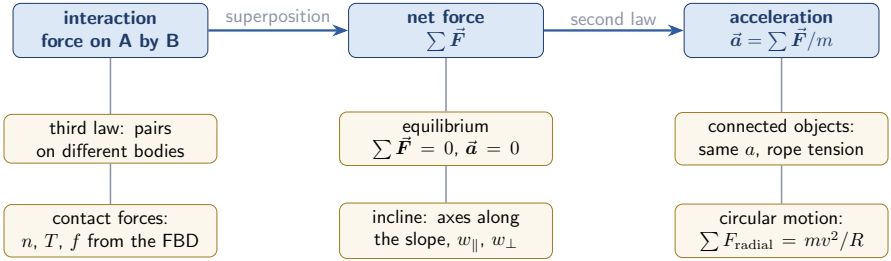
$T < m_2 g = 78.4\,\text{N}$; small a, because the pull barely exceeds the resistance. ✓
Reuse: Contact A: (3), (4); Contact B: (7), (8), (9).

Common mistakes and the correct habit

Mistake	Instead
Taking tension equal to a weight	T depends on a: use $\sum F = ma$ per body
Using $f_s = \mu_s n$ when not about to slide	$f_s \leq \mu_s n$; it adjusts
Friction opposes motion over the ground	It opposes sliding between surfaces
A "centripetal" or "centrifugal" arrow	Real forces supply the inward force
A separate "force down the slope"	It is a component of the weight
Swapping $\sin\theta$ and $\cos\theta$	Test $\theta = 0$ and $\theta = 90^\circ$
Letting mass decide $v_{\max}$ on a curve	$v_{\max} = \sqrt{\mu_s g R}$: mass cancels

Take away: static friction adjusts, kinetic friction is $\mu_k n$; the same laws apply everywhere; circular motion still obeys $\sum \vec{F} = m\vec{a}$; every hard problem starts with a free-body diagram.

Week 3 concept map



Key ideas

- **Every force is an interaction between two objects and has a named agent; a free-body diagram contains only the forces on the isolated body.**
- Net force is the vector sum of the forces on one object; mass is the resistance to acceleration, weight $w = mg$ is a force and depends on the place.
- $\sum \vec{F} = m\vec{a}$ **holds component by component; with** $\sum \vec{F} = 0$ **the velocity is constant (first law).**
- Choose axes along the acceleration or the surface; n comes from the perpendicular equation, not from $n = mg$.
- **Third-law pairs are equal and opposite and act on different bodies, so they never cancel.**
- Connected objects: one free-body diagram and one equation per body, linked by the constraint (same a, same T for an ideal rope).
- $f_s \leq \mu_s n$ is a limit and adjusts to the other forces; $f_k = \mu_k n$ is fixed; friction opposes sliding relative to the surface.
- **Uniform circular motion needs an inward net force** mv^2/R **supplied by real forces; in an inertial frame there is no "centrifugal" or "centripetal" arrow.**
- **Mantra:** isolate the body, draw the FBD, choose axes, apply $\sum \vec{F} = m\vec{a}$, check. Habit: sketch, predict, calculate, evaluate.

Reference tools: the next slides (master toolkit, core equations). The Reference deck holds the seven toolkits, the equation-anatomy slides and the master procedure.

• MASTER TOOLKIT – Force problems

1. IDENTIFY Which body or bodies, and what is asked?
2. ISOLATE One body at a time.
3. DRAW FBD(S) Forces on the body, each named by agent.
4. CHOOSE AXES Along the acceleration or the surface.
5. PREDICT Signs, direction, trend, rough size.
6. SELECT MODEL Equilibrium? Accelerating? Pair? Connected? Friction? Incline? Circular?
7. EXECUTE $\sum F = ma$ by components, one equation per axis per body; key move (right); algebra first, then numbers.
8. EVALUATE Units, sign, direction, limits, back to the FBD.

Result disagrees with the prediction? Investigate. Seven toolkits: Reference deck.

Step 7: the key move for each model

FBD: only forces on the body, each with an agent.
Axes: one axis along the acceleration or slope; resolve the rest.
Second law: $\sum F_x = ma_x$, $\sum F_y = ma_y$; n from the perpendicular equation; equilibrium is $a = 0$.
Third law: name, flip, locate; pairs act on different bodies.
Connected: one equation per body; same a and T.
Friction: sliding? $f_s \leq \mu_s n$ or $f_k = \mu_k n$.
Incline: $w_{\parallel} = mg\sin\theta$, $w_{\perp} = mg\cos\theta$.
Circular: $\sum F_{\text{radial}} = mv^2/R$ toward the centre; real forces only.

Framework: IDENTIFY (1–2), SET UP (3–6), EXECUTE (7), EVALUATE (8).

Reference tools: core equations

Idea	Equation	Conditions and units
Net force	$\sum \vec{\boldsymbol{F}} = \vec{\boldsymbol{F}}_1 + \vec{\boldsymbol{F}}_2 + \cdots$	vector sum; N
First law	$\sum \vec{\boldsymbol{F}} = 0 \Rightarrow \vec{\boldsymbol{a}} = 0$	inertial frame
Second law	$\sum F_x = ma_x, \quad \sum F_y = ma_y$	particle of constant mass; N
Third law	$\vec{\boldsymbol{F}}_{\text{A on B}} = -\vec{\boldsymbol{F}}_{\text{B on A}}$	different bodies
Weight	$w = mg$	$g = 9.8\,\mathrm{m/s^2}$ near Earth
Friction	$0 \le f_s \le \mu_s n, \quad f_k = \mu_k n$	μ dimensionless
Incline	$w_{\parallel} = mg\sin\theta, \quad w_{\perp} = mg\cos\theta$	θ from the horizontal
Circular	$\sum F_{\text{radial}} = mv^2/R = m\omega^2 R$	toward the centre
Flat curve	$v_{\max} = \sqrt{\mu_s g R}$	friction supplies the force

Where this goes next

Forces and the free-body diagram

The forces on your diagram decide which ones do work (Lecture 4), how momentum changes (Lecture 7), and which produce torque (Lecture 8).

Second law and energy

Energy methods (Lectures 4 and 5) give speeds without solving for a; the momentum form $\sum \vec{\boldsymbol{F}} = \mathrm{d}\vec{\boldsymbol{p}}/\mathrm{d}t$ is the general second law (Lecture 7).

Friction

Friction removes mechanical energy and warms the surfaces: work and energy in Lectures 4 and 5, heat and thermal energy in Lectures 9 to 11.

Inclines and ramps

The same slope problem is solved much more simply by energy conservation in Lecture 5, with no free-body axes.

Circular dynamics

$\sum F_{\text{radial}} = mv^2/R$ becomes the dynamics of rotating bodies in Lecture 8: torque and angular acceleration.

Physical meaning

The free-body diagram and component equations are the method you will keep: later topics change *which* forces appear, not how you set up the problem.

Checklist: can I . . . CORE

- ☐ name a force by who exerts it on whom?
- ☐ find a net force by components?
- ☐ distinguish mass, weight and normal force?
- ☐ draw a correct free-body diagram?
- ☐ apply the first and second laws by components?
- ☐ solve equilibrium problems?
- ☐ identify third-law pairs and explain why they do not cancel?
- ☐ solve connected-object problems with ideal ropes and pulleys?
- ☐ apply $f_s \le \mu_s n$ and $f_k = \mu_k n$ with the correct direction?
- ☐ solve incline problems with sensible axes?
- ☐ explain and apply $\sum F_{\text{radial}} = mv^2/R$?
- ☐ select the appropriate model in an unfamiliar problem?

Next

Tutorial sheet: the lift, extra FBD, connected-object, friction, incline and circular problems, the car that accelerates, two blocks joined by a rope, and the handcart whose friction changes the pull needed. The Reference deck holds the supporting material cut from the live lecture; the supplementary notes hold inertial frames, the momentum form of the second law, drag and terminal speed, and the four fundamental forces.

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